

CANONICAL LESSON

Census, Address, and Geospatial Frames

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C62
Notebook	V7_D_N12
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Build census, address, and geospatial frames with explicit coverage, currency, uniqueness, linkage quality, uncertainty, privacy, field verification, and controlled operational use.

Learning outcomes

Why this matters

Incomplete or duplicated frames can omit populations, distort allocation, expose households, and create false confidence in geographic coverage.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A statistical office reconciles address, building, imagery, and enumeration evidence before updating a census frame.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Diagnose duplicate addresses.
- 2.
- Separate building and dwelling.
- 3.

- Design a safe field-verification queue.

Answer key

- 1.
- Normalize components but preserve originals; combine spatial, name, and occupancy evidence with human review.
- 2.
- One building can contain many dwellings and households; identifiers and counts must respect each level.
- 3.
- Minimize sensitive attributes, use reason codes and safe routes, log corrections, and prohibit enforcement reuse.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.